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B.Tech. Degree IV Semester Examination April 2019

IT 1404 DATABASE MANAGEMENT SYSTEMS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain the characteristics of the database approach.
- (b) Explain the two types of data independence.
- (c) Compare spanned and unspanned records.
- (d) Write note on Heap files.
- (e) Explain relational model notation.
- (f) What is meant by normalization of data?
- (g) Explain schedule in transaction management.
- (h) List the features of data mining.

PART B

(4 × 15 = 60)

- II. (a) Explain the three schema architecture. (7)
- (b) Explain various DMBS interfaces. (8)
- OR**
- III. (a) Write short notes on : (10)
 - (i) Entity Types (ii) Entity Sets (iii) Attributes (iv) Keys
- (b) Explain Design Choices for ER Conceptual Design. (5)
- IV. Explain Different Hashing Techniques in detail. (15)
- OR**
- V. Briefly discuss the Index Structures for Files. (15)
- VI. Explain Basic Relational Algebra Operations. (15)
- OR**
- VII. (a) Explain the Inference Rules for Functional Dependencies. (8)
- (b) Briefly explain Boyce-Codd Normal Form. (7)
- VIII. (a) Describe the steps in building a datawarehouse. (10)
- (b) List ACID properties of Transaction. (5)
- OR**
- IX. (a) Illustrate with an example an algorithm for testing conflict serializability of schedule. (10)
- (b) Differentiate between serial and non serial schedule. (5)